

TANUSH KULKARNI

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science and Physics, Math Minor | James Scholar Honors Program

Expected: May 2028

GPA: 3.86/4.00

RELATED COURSEWORK

Data Structures

Computer Architecture

Classical Mechanics

Abstract Linear Algebra

Software Design Lab

Statistics and Probability II

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript/Typescript, Java, HTML, CSS, Kotlin

Frameworks/Tools: React, Node, Express, Flask, Git, GitHub, Pandas, Catch2, Postman/Bruno, Jest, scikit-learn, TensorFlow, OpenCV, Seaborn, PostgreSQL, MySQL, Docker, CI/CD, Mathematica

WORK EXPERIENCE

IDX Exchange

Remote

Software Development Engineer Intern

June 2026 – September 2026

- **Designed** and build a scalable, full-stack property search real estate application with a React frontend, Node/Express backend, and a MySQL database to store and access MLS property data.
- **Implemented** RESTful APIs, data-parsing pipelines for complex MLS datasets, and logging middleware to identify bottlenecks and improved database performance by an average of 99.2%.
- **Developed** a comprehensive test suite with Jest, including unit, and component tests, exceeding 70% test coverage on critical paths.

Center for Digital Transformation, Gabelli School of Business, Fordham University

Remote

Jr. Technical Member

May 2024 – Jan 2025

- **Developed** automated data aggregation systems using Selenium to collect and aggregate over 8000 data points from multiple databases, implementing filters to remove duplicates.
- **Performed** natural language processing on over 3000 documents, implementing semantic analysis, vectorization, and topic modeling to uncover patterns within documents, contributing to 3 published research papers.

PROJECT HIGHLIGHTS

Emotion Analysis Software (GitHub: <https://github.com/RollingTank/Emotion-Recognition>) Tools: **Python, AI, Vosk, Computer Vision**

- **Created** a robust emotion analysis software that assesses a user's emotional state by using speech transcription and the camera in real time and generates a report summarizing findings after the user closes the app while maintaining a CI/CD pipeline.
- **Implemented** multithreading in Python to simultaneously record video and audio, while ensuring a privacy-first design by only storing voice transcripts and not storing video recordings.

Charcot: AI Copilot for Psychiatric Assessments (Devpost: <https://devpost.com/software/charcot>) Tools: **React, Machine Learning**

- **Collaborated** with a team of four to develop an AI-powered assistant that implements real-time speech-to-text transcription using facial landmark detection with MediaPipe to help psychiatrists better understand their patient.
- **Extracted** behavioral metrics including eye contact monitoring, gaze stability tracking, and breathing rate estimation.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

Tutor (Previously Assistant Tutor)

Urbana, Illinois

CS 124, University of Illinois

August 2026 – Present

- **Guide** 50+ students, helping with coursework and student-driven projects, keeping students on track and providing feedback on projects.
- **Host** online tutoring office hours to assist students with course material, debugging code, and understanding fundamental computer science concepts, providing personalized support to students.

National Business and Engineering Organization (NOBE)

Urbana, Illinois

Technical Consultant

Sep 2025 – May 2026

- **Developed** RESTful API backend, implementing user authentication with password hashing and comprehensive data models for profiles, addresses, shopping carts, and order history.
- **Collaborated** on developing an internal attendance-tracking system for 190+ members, prioritizing reliability and detailed dashboards for administrators and regular users.